

Adaptive Invariant Computation: A Primer on Invariants, Recursive Compression and Reversion

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This is a public-facing background paper. It deliberately describes the research motivation and testable questions without disclosing implementation-specific details of the private Adaptive Invariant Computation architecture.

Abstract

Artificial intelligence systems increasingly rely on larger models, longer contexts and greater computational budgets. This primer introduces the research motivation behind *Adaptive Invariant Computation* (AIC): the idea that efficient computation may depend not only on compressing information, but on learning which structures must remain unchanged while compression proceeds. The approach grows from two earlier strands. The first is Bindu Recursive Compression/Consciousness Theory (BRCT), used here only as a conceptual model of recursive reduction toward stable structure. The second is *Inversion Is All You Need*, which proposed that an AI system should seek an invariant state beneath changing linguistic surface form before generating an answer. AIC asks a narrower engineering question: can a system progressively reduce its internal state while checking that task-relevant invariants remain intact, and stop before further reduction causes meaningful loss? This paper explains that question, relates it to information bottleneck methods, invariant learning, adaptive computation and reversible networks, and outlines a public experimental agenda. Detailed architectural mechanisms, training procedures and implementation-specific stopping logic are intentionally omitted pending further technical validation and IP review.

1 Why efficiency needs a preservation criterion

Compression is easy to state and difficult to trust. A shorter representation uses less memory, but it may also discard exactly the information that a later reasoning step needs. Larger context windows address this by retaining more information, but at a cost in memory, latency and computational complexity.

The central motivation for AIC is therefore simple:

Efficiency should not mean keeping less information indiscriminately. It should mean keeping less while preserving what still determines future correct behaviour.

This places the emphasis on invariants: properties that should remain stable under changes that do not alter the meaning or task outcome.

2 From surface form to invariant state

The earlier *Inversion Is All You Need* research direction argued that a language model should not rely only on the surface prompt it receives, but should first seek a deeper state that remains

stable across paraphrase, rhetoric and other superficial variation. The later Invariant-Space-LLM research draft explored this idea as a persistent internal workspace in a transformer [5].

A minimal conceptual form is

$$\text{observation} \rightarrow \text{invariant state} \rightarrow \text{response}. \quad (1)$$

The important point is not that language has one perfect hidden representation. Rather, many tasks contain relationships that should survive changes of wording, order or presentation.

3 What BRCT contributes

BRCT is an exploratory framework developed around recursive compression, convergence and a limiting stable centre referred to as a Bindu. Its broader consciousness interpretation is speculative and is not required to evaluate AIC.

For engineering purposes, only one motif is needed: repeated reduction can be imagined as a trajectory

$$z_0 \rightarrow z_1 \rightarrow z_2 \rightarrow \dots \quad (2)$$

that approaches a compact, stable state. The key research question is where to stop. If reduction proceeds too far, meaningful structure is lost. If it stops too early, the system retains unnecessary computational burden.

AIC therefore translates the Bindu metaphor into a practical question about a *boundary of safe reduction*: how far can a representation be simplified before task-relevant invariants are no longer preserved?

4 Why reversion matters

A system can appear stable while silently losing information. For this reason, AIC also takes inspiration from inversion and reversibility. Reversible neural networks have already shown that carefully structured transformations can reconstruct earlier activations and reduce memory requirements [4]. AIC asks a different but related question: after a lossy reduction, can the system still recover the *relevant structure* that was supposed to survive?

The public principle is therefore:

Compress, test what remained invariant, and test whether the relevant structure can still be recovered.

The precise mechanism used to perform those tests is part of the ongoing technical work and is not specified in this primer.

5 Geometric and phase-oriented background: FIT, GML, PPM and *Nanobrain*

A second line of background comes from Anirban Bandyopadhyay's work on Fractal Information Theory (FIT), Geometric Musical Language (GML), Phase Prime Metric (PPM), and the broader synthesis presented in *Nanobrain: The Making of an Artificial Brain from a Time Crystal* [?, ?]. These works investigate information through nested geometry, phase relations, timing and

non-linear structural organisation rather than treating computation only as a linear symbolic sequence.

The relevance to AIC is conceptual rather than architectural. FIT motivates thinking about information across nested scales; GML treats geometry and temporal relation as informational structure; PPM explores phase- and symmetry-oriented geometric relationships; and *Nanobrain* places those elements within a wider proposed physical artificial-brain framework. OpenGML subsequently provided an open-source environment for computational exploration of GML- and PPM-inspired nested geometry, phase and symmetry [?].

AIC does *not* assume that the brain is a time crystal, that PPM is a universal metric, that GML replaces conventional software, or that FIT replaces standard information theory. Instead, it extracts a narrower engineering intuition: a representation may contain relational structure that should remain stable even when its surface form changes. AIC translates that intuition into an independently testable question about whether preservation of task-relevant invariants can regulate representation reduction. In this sense, FIT/GML/PPM and *Nanobrain* are part of the intellectual lineage, while invariant preservation as an active computation-control criterion is the distinct AIC hypothesis.

6 Relationship to established machine learning

6.1 Information bottleneck

The information bottleneck framework seeks a compact representation that retains information relevant to a target variable [1]. AIC shares the concern with relevance-preserving compression, but investigates whether preservation can be monitored through a sequence of reductions rather than optimised only as a terminal representation objective.

6.2 Invariant learning

Modern self-supervised methods such as VICReg explicitly encourage agreement between representations produced from different views of the same underlying input while avoiding representational collapse [2]. This demonstrates that invariance can be made an explicit learning objective. AIC asks whether invariance can additionally guide the allocation and reduction of computational state.

6.3 Adaptive computation

Adaptive Computation Time allows a recurrent model to learn how many internal computation steps to take for different inputs [3]. AIC shares the broad idea that computation need not be fixed, but focuses on preservation of task-relevant structure as the reason to continue or stop reducing a representation.

6.4 Reversible computation

Reversible residual networks demonstrate that inversion can reduce memory by reconstructing activations rather than storing all of them [4]. AIC is not simply a reversible network. Reversion is instead treated as a conceptual test of whether relevant information survives a lossy transformation.

7 The AIC research question

The central public hypothesis can be stated without implementation detail:

A representation that is progressively reduced under an invariant-preservation constraint may achieve a better efficiency/robustness trade-off than a fixed compression strategy of similar capacity.

This is deliberately falsifiable. It could fail because invariants are difficult to learn, because the cost of testing preservation exceeds the savings from compression, because the system stops too early, or because conventional bottlenecks already provide the same benefit more simply.

8 A public experimental agenda

A credible evaluation should begin with controlled tasks where the transformations that preserve meaning are known. Candidate tests include symbolic problems under equivalent rewrites, relational structures under renaming or reordering, and semantic tasks under paraphrase or distractor insertion.

AIC should be compared against conventional learned bottlenecks, invariant-learning objectives and fixed-depth adaptive baselines. Evaluation should report not only task accuracy, but also representation size, memory requirement, computational cost and robustness under transformations not seen during training.

The research programme should also include negative controls in which the presumed invariant relationships are deliberately corrupted. If a method still appears to work when the invariance signal is meaningless, then the invariance mechanism is probably not responsible for the observed gain.

9 What this primer intentionally does not disclose

This paper does not specify:

- the internal form of the AIC preservation score;
- the detailed reversion mechanism;
- controller or stopping equations;
- training-loss composition or weighting;
- state-routing or compression policies;
- implementation-specific integration with transformer, memory or agent architectures.

These details are being treated separately while the technical approach is evaluated and any protectable foreground IP is identified.

10 Possible applications

If the underlying hypothesis is supported, the principle could apply to several forms of computational state rather than to one model type. Examples include compact latent representations, persistent agent memory, local and edge AI, selective context retention, structured retrieval, robotics and resource-constrained inference.

This breadth is also a reason for caution. An invariant-preserving system is not automatically safer or more truthful. It preserves whichever structure it has learned to treat as important. For safety-critical applications, invariant learning would have to be combined with explicit objectives, permissions, verification and external constraints.

11 BRCT, inversion and AIC: a clean separation

It is useful to keep the three levels distinct.

BRCT	Conceptual origin: recursive compression toward stable structure and a boundary beyond which the character of the state changes.
Inversion-first intelligence	Epistemic principle: seek the stable relational state beneath surface variation before generating from it.
AIC	Engineering hypothesis: use preservation of task-relevant invariants to govern how far internal state may be reduced.

This separation allows the engineering proposal to be tested independently of stronger philosophical claims. Evidence for or against AIC does not by itself prove or disprove BRCT as a consciousness theory.

12 Conclusion

Adaptive Invariant Computation begins with a simple challenge: how can an AI system know when it has compressed enough? The proposed answer is to make preservation of task-relevant invariants part of the computational decision itself. BRCT supplies the recursive-convergence intuition; inversion-first work supplies the focus on stable state beneath surface form; modern machine learning supplies the tools needed to make the resulting hypothesis measurable.

The next step is empirical. AIC should be judged by whether it can preserve performance and robustness while reducing representational or computational cost under matched, reproducible tests.

References

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